

CS 540 Midterm Sample Question Solutions

1. Which of the following sentence is FALSE regarding regression?

- (a) It discovers causal relationships
- (b) It relates inputs to outputs.
- (c) It is used for prediction.
- (d) It may be used for interpretation.

Answer: (a)

Explanation: (a) is *False*. Regression is a supervised learning algorithm. It can only predict values for inputs, which is not directly useful for discovering causal relationships. Note that (b) is true (regression finds a way to map inputs to outputs), (c) is true, since predictions can be made once the weights are learned, and (d) is true, as we can examine the outputs of regression on particular inputs to gain insights.

2. Consider the following statements regarding k -nearest neighbors.

I. If point A is among point B's k -nearest-neighbors, then B is among A's k -nearest-neighbors.

II. On a training set consisting of 100 positive items and 100 negative items, it is possible for 1NN to have training set accuracy 0 (i.e., it can completely fail on the training set).

- (a) I is True, II is False
- (b) I is False, II is True
- (c) Both are True
- (d) Both are False

Answer: (b)

Explanation: I: The statement is *False*. Consider a simple 1-D data set, A=0, B=1, C=5, D=6, E=7. Point C is among the 2-NN for point B (A and C are the nearest neighbors), but B is *not* among the 2-NN for C (Point D and E are the nearest neighbors)

II: The statement is *True*. It is always possible to construct a data set such that the nearest neighbor for each positive item is a negative item and vice versa. Hence, each item will be labeled the opposite and hence training set accuracy will be 0. You can visualize a simple 1-D data set with alternate positive and negative items.

3. What assumption does naive Bayes make?

- (a) Conditional independence of features

- (b) Conditional dependence of features
- (c) Independence of features
- (d) Dependence of features

Answer: (a)

Explanation: The statement in (a) is true: we assume the features are conditionally independent (when conditioned on the label). (b) is false, since we assume independence. (c) is false because the features will not be independent without conditioning on the label. (d) is false: the features are likely dependent, but this is not the key assumption for naive Bayes.

4. What is the primary difference between supervised and unsupervised learning? Please choose all that apply

- (a) Supervised learning gives you what some expected outputs are, unsupervised learning does not
- (b) There is very little difference, they are the same in most contexts
- (c) Supervised learning is often used for prediction tasks, but unsupervised learning is usually not
- (d) Unsupervised learning produces correct labels rather than being given correct labels

Answers: (a), (c)

Explanation: a) follows by definition. c) follows since unsupervised learning is often used to summarize data. Without given labels prediction is usually not doable. (b) is false, again by definition. (d) is false: unsupervised learning is not meant to produce labels at all typically.

5. Consider the following dataset about 5 types of mushrooms, about whether the mushroom is poisonous based on three binary features. We want to use the naive Bayes method to classify a new mushroom with Height=Tall, Body=Slim, and Color=Black. What's the probability of this mushroom being poisonous?

Height	Body	Color	Poisonous
Tall	Fat	Black	No
Tall	Fat	White	Yes
Short	Slim	White	No
Short	Slim	Black	Yes
Tall	Slim	White	Yes

- (a) 0.64
- (b) 0.82
- (c) 0.38
- (d) 0.67

Answer: (a)

Explanation: By Bayes' rule, we have

$$\begin{aligned}
 P(\text{Yes}|\text{Tall, Slim, Black}) &= P(\text{Yes, Tall, Slim, Black})/P(\text{Tall, Slim, Black}) \\
 &= \frac{P(\text{Tall, Slim, Black}|\text{Yes})P(\text{Yes})}{P(\text{Tall, Slim, Black}|\text{No})P(\text{No}) + P(\text{Tall, Slim, Black}|\text{Yes})P(\text{Yes})} \\
 &= \frac{\left(\frac{2}{3}\frac{2}{3}\frac{1}{3}\right)\left(\frac{3}{5}\right)}{\left(\frac{1}{2}\frac{1}{2}\frac{1}{2}\right)\left(\frac{2}{5}\right) + \left(\frac{2}{3}\frac{2}{3}\frac{1}{3}\right)\left(\frac{3}{5}\right)} \\
 &= 0.64.
 \end{aligned}$$

6. We are given data consisting of the following one-dimensional points 1, 3, 5, 6, 7, 12, 13, 15.

What will be the first pair of clusters in 2-mean clustering given the initial centers 1, 12?

- (a) {1, 3, 5, 6, 7}, {12, 13, 15}
- (b) {1, 3, 5, 6}, {7, 12, 13, 15}
- (c) {1, 3}, {5, 6, 7, 12, 13, 15}
- (d) {1, 3, 5, 6, 7, 12}, {13, 15}

Answer: (b)

Explanation: Self-explanatory. 1,3,5,6 are closer to 1 than to 12, and the reverse for the remaining points.

7. We have the following one dimensional dataset that has been merged into three clusters: $C1 = \{1, 2\}$, $C2 = \{4, 5\}$, $C3 = \{6, 10\}$. Which clusters will be merged next using single linkage? Which will be merged using complete linkage?

- (a) C2 and C3, C1 and C2
- (b) C1 and C2, C2 and C3
- (c) C1 and C3, C2 and C3
- (d) C2 and C3, C1 and C3

Answer: (a)

Explanation: Using single linkage, C2 and C3 have a minimum distance of $|5 - 6| = 1$.

This is the minimum distance between all clusters, so they are merged. Using complete linkage, C1 and C2 have a maximum distance of $|1 - 5| = 4$. This is the minimum distance between any two clusters, so C1 and C2 are merged.

8. Suppose we have the following training corpus in an NLP setup. “I took my dog out for a walk but my dog ran away” (i) What is the $P(\text{dog})$ using a unigram model without Laplace smoothing? (b) What is the $P(\text{my dog})$ using a unigram model without Laplace smoothing? (c) What is the $P(\text{my dog})$ using a bigram model without Laplace smoothing?

Please show your work for each part.

Solution:

(a) $P(\text{dog}) = \frac{2}{13}$ (note there are 13 words in the sentence; two are ‘dog’).

(b) $P(\text{my dog}) = P(\text{my}) * P(\text{dog}) = \frac{2}{13} * \frac{2}{13} = \frac{4}{169}$

(c) $P(\text{my dog}) = \frac{2}{12} = \frac{1}{6}$

9. Suppose you are performing PCA in $\mathbb{R}^3$ (3-dimensional space). It turns out that the first

principal component is $v_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ and the second principal component is $v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$. One

of the original data points is $x = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$. What will its new representation be after projection?

Solution: Just multiply onto the two components for the projection. We get the representation $v_1 = \begin{bmatrix} \frac{6}{\sqrt{2}} \\ \frac{10}{\sqrt{2}} \\ \frac{10}{\sqrt{2}} \end{bmatrix}$. Note that since we are using two principal components, our new representation should be 2-dimensional.

10. You have a dataset for regression given by $(x_1, y_1) = \left(\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, 2 \right)$, $(x_2, y_2) = \left(\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, 4 \right)$.

(a) What are the labels? What is the number of points n and the dimension of the feature vectors d ?

(b) We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. Predict $\hat{y}$ for $x = \begin{bmatrix} 1 \\ 10 \\ 1 \end{bmatrix}$.

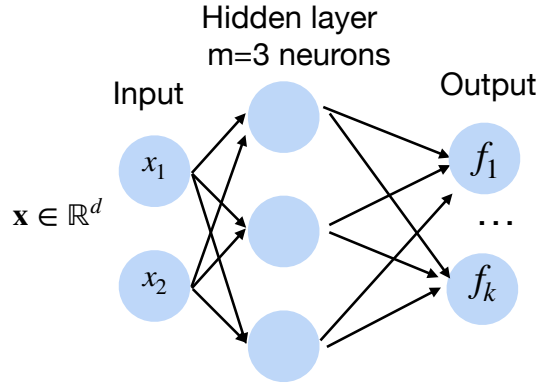
(c) For the weights above, computed the mean squared error.

Solution: (a) The labels are $y_1 = 2$ and $y_2 = 4$. There are $n = 2$ points and $d = 3$.

(b) The prediction is just $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 = 2(1) + 1(10) + 1(1) = 13$.

(c) The MSE is $1/2((-1 - 2)^2 + (8 - 4)^2) = 1/2(9 + 16) = 12.5$.

11. Suppose we want to solve the following k -class classification problem with cross entropy loss $\ell(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{j=1}^k y_j \log \hat{y}_j$, where the ground truth and predicted probabilities $\mathbf{y}, \hat{\mathbf{y}} \in \mathbb{R}^k$. Recall that the softmax function turns the output into probabilities: $\hat{y}_j = \frac{\exp f_j(x)}{\sum_i^k \exp f_i(x)}$. A diagram is provided below. What is the partial derivative $\partial_{f_j} \ell(\mathbf{y}, \hat{\mathbf{y}})$?



For notational simplicity, we use y_i to denote $\mathbf{1}\{y_i = 1\}$, and $\hat{y}_i$ as $p(y_i = 1|\mathbf{x}; \theta)$.

Solution: Rewrite

$$\begin{aligned} \ell(\mathbf{y}, \hat{\mathbf{y}}) &= -\sum_{j=1}^k y_j \log \frac{\exp(f_j)}{\sum_{i=1}^k \exp(f_i)} \\ &= \sum_{j=1}^k y_j \log \sum_{i=1}^k \exp(f_i) - \sum_{j=1}^k y_j f_j \\ &= \log \sum_{i=1}^k \exp(f_i) - \sum_{j=1}^k y_j f_j. \end{aligned}$$

We have $\partial_{f_j} \ell(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\exp(f_j)}{\sum_{i=1}^k \exp(f_i)} - y_j = \hat{y}_j - y_j$.